INTRODUCTION

We have already learnt in the previous unit that the transition metals have a tendency to form complexes (**coordination compounds**). The name is derived from the Latin words *'complexus'* and *'coordinate'* which mean *'hold'* and *'to arrange'* respectively. The complexes of transition metals have interesting properties and differ from simple ionic and covalent compounds. For example, chromium(III)chloride hexahydrate, $CrCl_3.6H_2O$, exists as purple, pale green or dark green compound. In addition to metals, certain non metals also form coordination compounds but have less tendency than d block elements. Coordination compounds play a vital role in the biological functions, and have wide range of catalytic applications in chemical industries. For example, haemoglobin, the oxygen transporter of human is a coordination compound of iron, and cobalamine, an essential vitamin is a coordination compound of cobalt. Chlorophyll, a pigment present in plants acting as a photo sensitiser in the photosynthesis is also a coordination compound. Various coordination compounds such as Wilkinson's compound, Ziegler Natta compound are used as catalysts in industrial processes. Hence, it is important to understand the chemistry of coordination compounds. In this unit we study the nature, bonding, nomenclature, isomerism and applications of the coordination compounds.

5.1 Coordination compounds and double salts:

When two or more stable compounds in solution are mixed together and allowed to evaporate, in certain cases there is a possibility for the formation of double salts or coordination compounds. For example when an equimolar solution of ferrous sulphate and ammonium sulphate are mixed and allowed to crystallise, a double salt namely Mohr's salt (Ferrous ammonium sulphate, $FeSO_4.(NH_4)_2SO_4.6H_2O$) is formed. Let us recall the blood red colour formation in the inorganic qualitative analysis of ferric ion, the reaction between ferric chloride and potassium thiocyanate solution gives a blood red coloured coordination compound, potassium ferrithiocyanate $K_3[Fe(SCN)_6]$. If we perform a qualitative analysis to identify the constituent ions present in both the compounds, Mohr's salt answers the presence of Fe^{2+} ,NH_4^+ and SO_4^{2-}ions, whereas the potassium ferrithiocyanate will not answer Fe^{3+} and SCN^- ions. From this we can infer that the double salts lose their identity and dissociates into their constituent simple ions in solutions , whereas the complex ion in coordination compound, does not loose its identity and never dissociate to give simple ions.

5.2 Werner's theory of coordination compounds:

Swiss chemist Alfred Werner was the first one to propose a theory of coordination compounds to explain the observed behaviour of them.

Let us consider the different coloured complexes of cobalt(III) chloride with ammonia which exhibit different properties as shown below.

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